Structured Outputs Do Not Imply Compatible Uncertainty: An Empirical Study of Confidence Signals in a Multi-Step Decision Workflow

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Abstract

Structured-output decision APIs return well-typed answers together with numeric confidence, and it is tempting to treat those numbers as interchangeable probabilities that can be composed across the steps of a workflow. We examine that practice for the three decision primitives of one commercial API: a categorical `choice`, a binary `noul` and an ordinal `score`. The primitives are chained into a three-step triage workflow (route, assess, rate severity). We evaluate on controlled constructed items, on 202 Mozilla bugs labeled by triage and on 240 Linux kernel bugs labeled by their reporters, using 4,584 API calls in the main real-data runs. Three findings hold on both trackers. First, the primitives' confidence signals have distinct failure modes. `choice` confidence is saturated (85–96% of real bugs at ≥ 0.90) yet discriminative (AUROC 0.90 [0.83, 0.96] and 0.98 [0.95, 1.00]). `noul`'s confidence-like value is uninformative (AUROC 0.53 [0.42, 0.65] and 0.41 [0.32, 0.50]). `score`'s confidence is flat across true severity levels. Second, fixed confidence thresholds designed on constructed data lose their intended selectivity on real text: a cut that escalated 73% of constructed items (target 67%) escalated 88% and 78% of real bugs (targets 41–42%). Third, `score` compresses severity toward the middle level, consistent with central-tendency bias reported for LLM raters in other domains, and its confidence gives no indication of it. In exploratory analyses, the composition rules we tested did not consistently identify which cases to escalate: ranking escalations by composed confidence did no better than random selection on either tracker (one borderline worse-than-random estimate did not replicate), and the intervals cannot exclude a modest signal. We also show that a natural comparison of confidence-based against decision-based composition is confounded when the target action is itself a function of the decisions. The corrected comparison is pre-registered for a controlled follow-up that has not yet been run.

1 Introduction

A model can return a perfectly valid, well-typed structured response and still make the wrong decision. When several such decisions are chained into a workflow, errors at one step propagate to the next. A common hope is that each step's reported confidence can be composed into a workflow-level signal that says when to hand a case to a human.

We ask a narrower and more testable question. *When a workflow chains heterogeneous decision primitives, can it rely on their uncertainty signals, and how should it combine them?* Heterogeneity matters because the API we study exposes three primitives whose uncertainty is *structurally* different. `choice` returns a distribution over labels plus a scalar confidence. `score` returns an ordinal distribution plus a scalar confidence. `noul` returns a single probability with no separate confidence field. Treating these numbers as one kind of quantity is a modeling assumption, and this paper tests it.

Contributions.

1. **Characterization** (Section 4). Per-primitive behavior of confidence on controlled and real data, showing three incompatible profiles.
2. **Transfer** (Section 5). Fixed thresholds set on constructed content lose their selectivity on real text, in two distinct ways.
3. **Real-data validation** (Section 6). Two bug trackers from different projects, with different labeling processes (triage and reporter), on which severity compression appears in both.
4. **Composition** (Section 7, exploratory). The confidence-based escalation rules we tested did not consistently pick better cases than random on either tracker. We identify, and correct for the follow-up, a confound in the obvious comparison.
5. **Method** (Section 8). A pre-registration and audit process that caught six design or implementation defects, most of them before any evaluation data existed.

Non-claims. We make no claim about general calibration, other APIs or model versions, domains beyond software bug triage, or real-world operational benefit.

2 Related work

Calibration of single decisions. Post-hoc calibration [20, 21] and the miscalibration of modern networks [14] are well studied, and confidence estimation for language models has been surveyed in [13].

Selective prediction and deferral. Risk–coverage trade-offs and reject options [10–12], learning to defer [19] and prediction sets in human–AI teams [4] address *one* decision.

Conformal methods and composition. Conformal prediction [2] and conformal risk control [3] give distribution-free guarantees. Composition across stages has been studied for two stages [28] and for multi-stage NLP and LLM pipelines, where errors compound across stages and independently calibrated stages lack a joint coverage guarantee [17]. Error propagation through pipeline steps has also been quantified [7]. Classical reliability theory composes component reliabilities [5].

Cascades and escalation. Cascades route a *single* query through model tiers. Prior work studies early abstention [25], the optimal depth and deferral thresholds of cascades [6], conformal cascades [8], confidence tuning [22] and judges with escalation guarantees [15].

Combining confidence across components. Fixed combination rules such as the product rule are classical [16]. The product rule's usual justification assumes conditional independence and comparable probability scales, and these are precisely the assumptions our strategy S2 makes implicitly. At the system level, a recent taxonomy argues that uncertainty in LLM applications must be studied across the whole compound system, as it propagates within models, across system components and through socio-technical processes [24]. Cost-sensitive deferral to multiple experts under workload limits [1] is closely related to the budgeted escalation setting our follow-up uses.

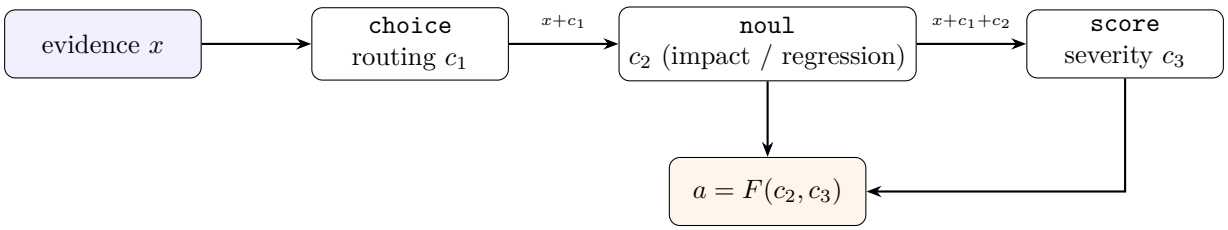


Figure 1: The three-step workflow. Each node is called three times. The action rule F uses only c_2 and c_3.

Ordinal outputs and central tendency. LLM raters have been reported to compress ordinal scores toward the middle of the scale, for example in clinical scoring, where prompt changes did not remove the effect [26]. Few-label calibration of ordinal LLM outputs models such flattening explicitly [23]. Our severity result is consistent with this literature. What we add is that it appears in bug triage, against two labeling processes, and that the primitive's own confidence gives no sign of it.

Agents and trajectories. Closest to us, a counterexample shows that step-level calibration need not imply trajectory-level calibration for LLM agents, and step errors are empirically coupled along trajectories [18]. Related work calibrates agent trajectories [27] and studies uncertainty propagation in multi-step agentic decisions [9].

Positioning. Prior work studies calibration of single decisions, cascades of the *same* decision, agent trajectories, and system-level propagation in general. Our contribution is narrower and empirical. We measure how the uncertainty signals of three *structurally different* primitives of one API behave inside one workflow against real labels, and whether simple composition rules over them identify escalation candidates. We are not aware of prior work with this specific design. Our literature search was web-based and did not systematically cover reliability engineering or HCI escalation research, so we make no stronger priority claim.

3 Setting

Primitives and repeats. All runs use model version `jev-1.13.0`. Each node is called three times, and the three answers are reduced to one decision. For `choice` and `score` we take the modal decision with a tie tolerance of 0.02; a node with no two-of-three majority is *repeat-unstable*. For `noul` we threshold the mean probability at 0.5, and its confidence-like value is $2\,|\bar{p} - 0.5|$.

Workflow. Node 1 (`choice`) routes the incident to a network or storage subsystem. Node 2 (`noul`) assesses a binary property: customer impact on constructed data, regression on real bugs. Node 3 (`score`) rates severity on three levels. Each node reads the evidence plus the upstream decisions, appended by a fixed adapter (Fig. 1).

Target action. The target is $a^* = F(c_2^*, c_3^*)$, where F maps an *unclear* c_2 to *request more evidence*, $c_3 = 2$ to *escalate*, $c_3 = 1$ with $c_2 = 1$ to *defer*, and everything else to *accept* (Table 6). F is a construct we defined, not an independently validated outcome, and every strategy result is conditional on it.

	Constructed items	Mozilla (n=202, triage)	Kernel (n=240, reporter)
`choice`	High-confidence error subgroup: mean confidence 0.915, accuracy 0.333 (n=6)	85% at ≥ 0.90, agreeing 0.96; AUROC 0.90 [0.83, 0.96]; ECE 0.05	96% at ≥ 0.90, agreeing 0.99; AUROC 0.98 [0.95, 1.00]; ECE 0.02
`noul`	24/24 correct, confidence below accuracy in every bin	Agreement flat across bands (0.71–0.84); AUROC 0.53 [0.42, 0.65]; ECE 0.36	Non-monotonic (0.58–0.93); AUROC 0.41 [0.32, 0.50]; ECE 0.36
`score`	Tracks the most severe sentence (6/6); confidence varies by 0.173 with sentence order alone (1 of 3 items)	Confidence ≈0.6 at every true level; AUROC 0.58 [0.50, 0.66]	Confidence 0.65 / 0.62 / 0.61 by true level; AUROC 0.57 [0.50, 0.64]

Table 1: Per-primitive behavior of the confidence signal. AUROC is for predicting the node's own agreement with the reference label, with 95% bootstrap intervals.

Strategies.

- **S1:** F applied to the node decisions.
- **S4a:** S1 plus escalation on repeat instability.
- **S2:** escalate the $\lceil qn \rceil$ instances with the lowest product of per-node confidences and accept the rest. Here q is the share of non-accept targets.
- **S5:** the same, ranked by per-primitive rank-normalized uncertainty.
- **2h / 5h:** S1 plus confidence-ranked escalation of a fixed 10% budget.
- **R:** the same budget escalated at random.

Statistics. Agreement rates carry Wilson 95% intervals. AUROCs use the Mann–Whitney estimator, with 95% percentile bootstrap intervals over bugs ($B = 2{,}000$, fixed seed). Bugs are distinct reports with no shared unit, so we resample bugs. Paired strategy comparisons use the exact McNemar test on discordant pairs, and the difference in $P(Y_{\text{decision}})$ gets a bootstrap interval. Within every resample, the set-relative rankings (S2, 2h) are recomputed. We compare the two trackers through the difference of their paired differences, with independent bootstraps per tracker. We do not compare their p-values. For budgeted overrides, the reference is the *exact* expected cost of k random overrides. The randomization p-value (10,000 random k-subsets) is conditional on the evaluated bugs, and a bootstrap interval over bugs gives the sampling uncertainty. An interval that covers the null value is read as *no evidence of an effect*, not as evidence of no effect. The scripts are `paper_statistics.py` and `mozilla_exploratory_analysis.py`.

Outcomes. Outcomes are $P(Y_{\text{decision}})$, the probability that the action equals a^*, and the cost under a frozen matrix (Table 7), in which a wrong *accept* costs 6–10, any other mismatch 1, and each escalation an extra 1 for human time.

4 Characterization: incompatible uncertainty semantics

Table 1 shows three incompatible profiles. `choice`'s confidence is almost always near 1.0. It still discriminates, because the rare low-confidence answers are where its errors are. The high-confidence

Rule (designed on constructed data)	Constructed pilot	Mozilla	Kernel
Escalate if `choice` confidence ≥ 0.90 (share fired)	27/30	85%	96%
Accept iff confidence product ≥ 0.5 (share escalated)	73%	88%	78%
target escalation share	67%	41%	42%

Table 2: Selectivity of the two fixed-threshold rules, measured on the rules' own input signals.

	Mozilla		Kernel	
Node	Agreement [95% CI]	Baseline	Agreement [95% CI]	Baseline
`choice`	0.89 [0.83, 0.92]	0.59	0.975 [0.95, 0.99]	0.50
`noul`	0.80 [0.74, 0.85]	0.74	0.725 [0.67, 0.78]	0.81
`score`	0.53 [0.46, 0.60]	0.40	0.44 [0.38, 0.50]	0.33

Table 3: Node agreement with reference labels (Wilson intervals). Each baseline is the most frequent label.

error subgroup found on deliberately ambiguous constructed items does not recur on real text (2 of 6 kernel errors are at ≥ 0.90). `noul`'s confidence-like value carries no usable information about correctness on either tracker. Its Mozilla interval covers 0.5, and its kernel interval lies at or just below 0.5 (weakly anti-informative). `score`'s confidence does not vary with the true level, so it gives no warning of the systematic bias in Section 6. Any composition rule that multiplies or averages these numbers inherits this mix.

5 Fixed thresholds lose their selectivity on real text

The rules in Table 2 fail in two different ways. The `choice` trigger is non-selective *everywhere*, because `choice` confidence is saturated: that is a design failure, visible only once measured. The product cut is *distribution-sensitive*. It was near its target on constructed data and far from it on real data, because real text pulls `noul` toward 0.5 and so pulls the product down. A fixed threshold carries the confidence scale of the data it was set on, and our constructed data was too clean to reveal this. The two Revision-10 strategies built on these thresholds scored 0.37–0.40 (Mozilla) and 0.34–0.37 (kernel). We replaced them with rank-based rules at a matched escalation rate before any evaluation.

6 Real-data validation on two trackers

Data. *Mozilla:* 202 fixed Firefox networking and storage bugs. Severity S1–S4 is assigned by triage and mapped to levels 2/2/1/0, and the regression flag is a keyword. Bugs whose severity was revised were excluded. *Linux kernel:* 240 fixed networking and storage bugs. Severity (blocking/high/normal/low $\rightarrow$ 2/2/1/0) is set by the reporter, and the regression flag is a dedicated field. Severity is rarely changed after filing: 15 of the 240 sampled bugs, all moving away from the default "normal". Both samples are balanced at 40 per (subsystem, level) cell where available. Texts are cleaned, privacy-scrubbed and truncated to 2,000 characters. Agreement is with the reference label, not ground truth.

Routing and regression. `choice` agrees with the component label well above baseline on both trackers (Table 3). It still agrees 0.84 on Mozilla texts with no subsystem keyword. `noul`'s errors

	Mozilla: `score` output			Kernel: `score` output		
True level	0	1	2	0	1	2
0 (minor)	**0.21**	0.66	0.13	**0.25**	0.68	0.08
1 (normal)	0.09	**0.72**	0.19	0.06	**0.65**	0.29
2 (serious)	0.01	0.41	**0.57**	0.01	0.55	**0.44**

Table 4: Row-normalized confusion of `score` against reference severity (repeat-unstable outputs excluded). Mass concentrates on the middle level.

change direction between trackers. On Mozilla it mostly misses regressions (23 misses vs. 17 false alarms). On the kernel it over-flags them (63 false alarms vs. 3 misses), which leaves it below the always-"no" baseline. The kernel false-alarm rate is the same on bugs explicitly marked "not a regression" (33/100) as on bugs with the field unset (30/94). So unset labels do not explain it, and on bugs with a set field `noul` beats its baseline (110/146 vs. 100/146).

Severity compression. Table 4 shows the row-normalized confusion of `score`. On both trackers, most minor bugs are rated higher than their label: 79% [0.67, 0.88] on Mozilla and 75% [0.65, 0.83] on the kernel. A large share of serious bugs are rated lower: 43% and 56%. The middle level absorbs both. With Mozilla's own severity definitions in place of ours, serious bugs improve but normal bugs are pushed up, and agreement moves only from 0.53 to 0.56. The criteria matter, but the compression persists. The two trackers differ in project, bug population and labeling process. They share the API, the prompts, our sampling design and, plausibly, training-data exposure, so the replication speaks against an artifact of one *labeling* process, not against every shared factor. The pattern matches central-tendency bias reported for LLM raters in other domains [26].

7 Composition (exploratory)

All analyses in this section were specified after the Mozilla results had been seen. They are hypothesis-generating, and the confirmatory test is pre-registered (Section 10).

The naive comparison is confounded. On Mozilla, decision composition (S1) beats confidence composition (S2) by 0.17 [0.08, 0.27] in $P(Y_{\text{decision}})$, with 61 vs. 27 discordant pairs (exact McNemar $p = 0.0004$). But S2 never reads the node decisions, while a^* escalates on the severity *decision*. To be right on a serious bug, S2 needs low confidence to *mean* high severity, so the comparison mostly measures whether confidence can stand in for severity. On the kernel the difference is smaller and inconclusive: 0.06 [−0.03, 0.15], 66 vs. 51 discordant pairs, $p = 0.20$. The difference between trackers is 0.11 [−0.01, 0.24]. Its interval covers zero, so we have *no evidence that the effect differs* between trackers, only a weaker estimate on one of them.

The fair question, and two different targets. Does confidence add anything *beyond* the decisions? Starting from the decision-based action F, we also escalate the k least-confident bugs that F did not escalate, for every k, and compare with the exact expectation under random selection (Fig. 2). We report two AUROCs with different prediction targets, which should not be conflated:

- **(A) Missed serious bugs.** Among bugs that F did not escalate, does low composed confidence rank those whose target is *escalate* first? This is what an override budget can fix. AUROC 0.49 [0.37, 0.61] on Mozilla and 0.50 [0.40, 0.60] on the kernel.

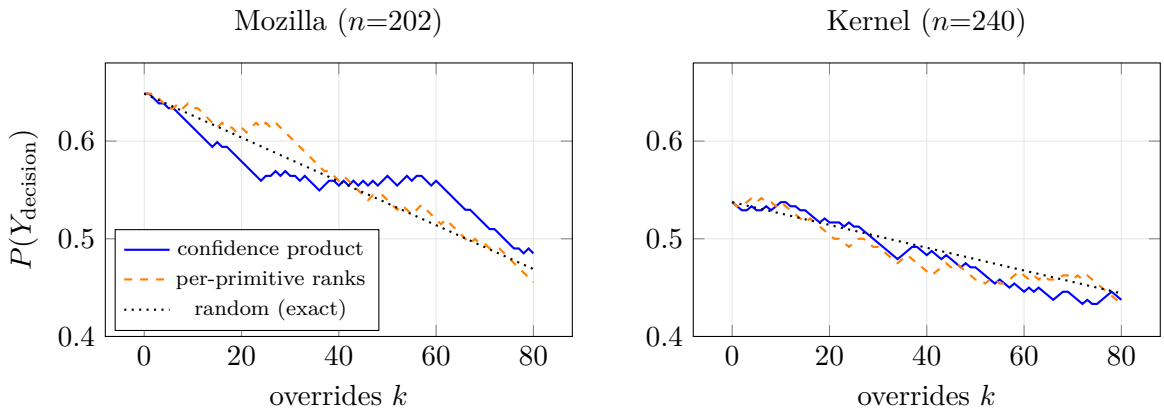


Figure 2: Adding confidence-ranked escalations on top of decision composition. At $k = 0$ the curve is S1. Neither ranking consistently improves on random selection of the same number of cases: deviations are a few points at most and change sign between rankings and trackers. For example, per-primitive ranks run slightly above random on Mozilla for $k \approx 15$–35, and the confidence product runs below.

- **(B) Any decision error.** Across all bugs, does low composed confidence rank those where F's action is wrong first? AUROC 0.54 [0.45, 0.63] on Mozilla and 0.47 [0.40, 0.54] on the kernel.

Every interval covers 0.5. The data show no evidence that composed confidence ranks either target better than chance, but they cannot rule out a modest signal (AUROC up to about 0.6).

Budgeted overrides, scored by cost. At a fixed budget of $k = \lceil 0.1n \rceil$ overrides ($k = 21$ on Mozilla, 24 on the kernel), let T be the total cost of the confidence-ranked overrides and μ the exact expected total under k overrides drawn uniformly from the same candidates. Two analyses answer two different questions.

- **Randomization test, conditional on the evaluated bugs.** This asks whether the ranking selected worse or better overrides than random selection *on these bugs*. The statistic is $|T - \mu|$. The null distribution comes from 10,000 random k-subsets of the candidates, and the test is two-sided.
- **Bootstrap over bugs.** This asks how stable the per-bug difference $(T - \mu)/n$ is under resampling of the bugs, with rankings and μ recomputed in each resample.

On Mozilla, $T = 390$ against $\mu = 357.2$. The observed value sits at the edge of the null distribution (2.5–97.5% range 323–388), so $p = 0.05$. The per-bug difference is +0.16 [−0.01, 0.24], and 97% of resamples are positive. That is borderline evidence that the product ranking chose *worse* overrides than random on Mozilla. On the kernel, $T = 338$ against $\mu = 358.6$ ($p = 0.18$), a per-bug difference of −0.09 [−0.23, 0.06]: inconclusive, and in the opposite direction. These analyses are exploratory and uncorrected for multiplicity. We read them as no consistent benefit from confidence-ranked overrides, with one borderline harmful estimate that did not replicate, and not as an established harm.

Defect	Detected by	When
A "primitive-aware" arm structurally identical to the baseline	inspecting the action rule	before evaluation
A trigger firing on ≈90% of inputs	selectivity on input signals only	before evaluation
A threshold sensitive to the data distribution	smoke test vs. pilot	before the full run
Four silent runner defects (e.g. the literal "None" shown to the model; partial data analyzed)	adversarial probing	before the live run
The confidence-vs-decisions confound	post-hoc review	after the Mozilla result
Template content whose labels were recoverable from punctuation alone (100%) and whose sentences leaked cross-label cues	independent review (separate AI agent)	before the controlled study

Table 5: Defects found by the study's own audits.

A structural reason. In this workflow the only discriminative confidence (`choice`, AUROC 0.90–0.98) belongs to the one node that F does not use. The informative signal never reaches the action, and the two signals that do reach it are close to uninformative.

8 Method: audits that caught defects

Table 5 lists the defects the study's own audits caught. Most were found before the data that would have exposed them existed. The exception is the confound, found after a result. It would have been caught earlier by asking, for every strategy and every target class: *can this strategy be right here, and through what information?* We recommend this check, together with measuring each trigger's selectivity on pilot inputs, before freezing any escalation policy.

9 Limitations

- **Generality.** We study one API at one model version, in one domain.
- **References are not truth.** Mozilla's triage sample excludes re-triaged bugs, which biases agreement upward. Kernel labels are reporters' claims, and the kernel's middle level is a default value.
- **Small constructed samples.** The `choice` error subgroup is $n = 6$, and `noul`'s perfect accuracy on constructed data is 24 items.
- **Constructed target.** $a^* = F(\cdot)$ is constructed, and all strategy results are conditional on it.

- **Memorization.** Old public bugs may appear in training data. The recent subsets are small (Mozilla 54, kernel 35).
- **Protocol revisions.** The protocol went through twelve revisions. All are dated and disclosed, and each was made before the data it affects existed. The number is nonetheless large.

10 Pre-registered follow-up (planned; not yet run)

No result in this paper comes from this study. To test the composition question under control, we pre-registered a study on deterministic template content: 3,103 units, each instance describing one incident about one service, with value-neutral sentence frames and 13 automated audits. Its primary tests are H1, confidence-ranked vs. random overrides at a 10% budget scored by cost, and H2, per-primitive vs. raw-product ranking, with a sign-flip test clustered by service. Both are Holm-corrected. The sample size is set from a second calibration pilot, with a content ceiling of 720 instances. Given the results above, a null H1 is a live outcome, and it would be reported as the answer to the composition question. Deterministic templates raise their own external-validity questions. That is why the real-data studies above are reported separately, and why their conclusions do not depend on the follow-up.

11 Conclusion

Structured outputs make uncertainty *look* uniform: every primitive returns a number between 0 and 1. In the evaluated workflow, the confidence values of three heterogeneous primitives showed distinct failure modes. One was saturated but informative, one was uninformative, and one was flat while its decisions were systematically biased. Fixed thresholds on them did not keep their selectivity across content distributions, and the composition rules we tested did not consistently identify cases for escalation. Severity compression appeared on two bug trackers with different labeling processes and was invisible to the primitive's own confidence. These findings do not show that confidence composition fails in general. They show that primitive-level confidence values should not be assumed to be directly composable, and that uncertainty should be evaluated at the level of the workflow that uses it.

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A Action rule and cost matrix

$c_2 \setminus c_3$	0	1	2
1 (yes)	accept	defer	escalate
0 (no)	accept	accept	escalate
unclear	request evidence	request evidence	request evidence

Table 6: The action rule $F(c_2, c_3)$. *Unclear* is checked first.

action \ target	accept	escalate	request evidence	defer
accept	0	10	6	8
escalate	1	0	1	1
request evidence	1	1	0	1
defer	1	1	1	0

Table 7: Frozen cost matrix $\text{cost}(a, a^*)$. One unit of human cost is added per escalation.

B Reproducibility

All calls used `jev-1.13.0`. Real-data calls: Mozilla 2,424 (including a label-wording arm) plus a 252-call smoke test; kernel 2,160 plus a 216-call smoke test. All returned HTTP 200 and parsed. Every dataset, raw response, execution manifest and analysis is hash-locked in the accompanying repository, as are the pre-registration documents and their revision history. The figure data in this paper are generated from committed result files by `make_figure_data.py`.